

AN13408

Smart Kitchen Demo

Rev. 0 — 18 October 2021

Application Note

1 Introduction

This application note discusses how to load and operate the Smart Kitchen Demo for i.MXRT1050 and covers some key aspects of the demo. This application uses i.MXRT1050 and the NXP GUI Guider graphical user interface development tool. To run the associated code and view the GUI Guider source, you need:

- GUI Guider version 1.1.0 or later (available at www.nxp.com/gui-guider)
- MCUXpresso version 11.3.0 or later
- MIMXRT1050-EVKB evaluation kit
- i.MXRT1050 SDK package
- Rocktech RK043FN02H-CT LCD display

2 Smart Kitchen demo overview

The Smart Kitchen Demo consists of three main parts: an Oven portion, a Coffee Machine portion, and a Blender portion.

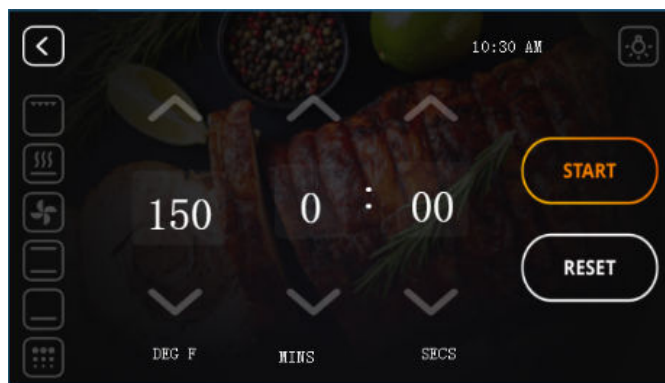


Figure 1. Oven portion

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Figure 2. Coffee Machine portion



Figure 3. Blender portion

Each demo portion is an independent demonstration of a part of a smart kitchen. You can navigate back to the start screen from any of the demos, a restart of the device is not required to run a different demo.

2.1 Loading the Smart Kitchen demo

This section gives detailed information about loading the Smart Kitchen demo.

2.1.1 Loading the Smart Kitchen Demo into MCUXpresso IDE

1. To load the Smart Kitchen Demo into MCUXpresso IDE, first open MCUXpresso IDE. It is recommended (but not required) to choose an empty folder for the workspace.

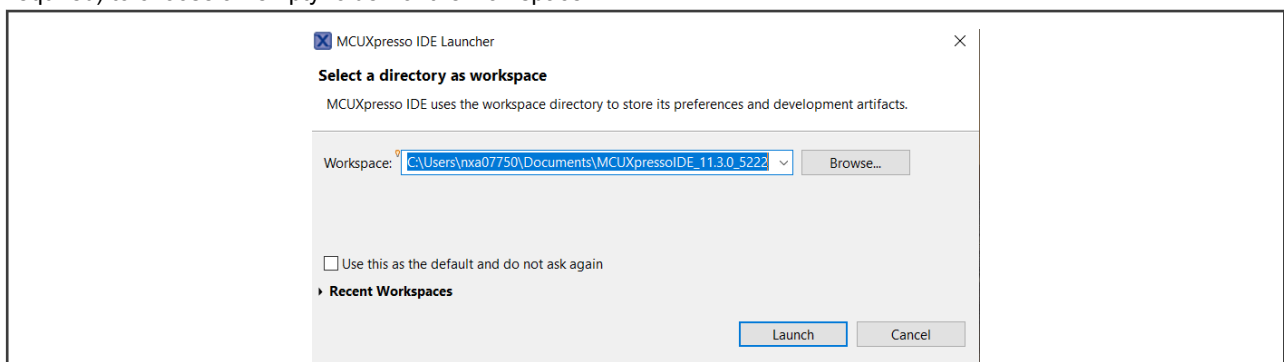


Figure 4. Selecting workspace in the MCUXpresso IDE Launcher

2. Once the workspace is loaded, select **File-> Import**.

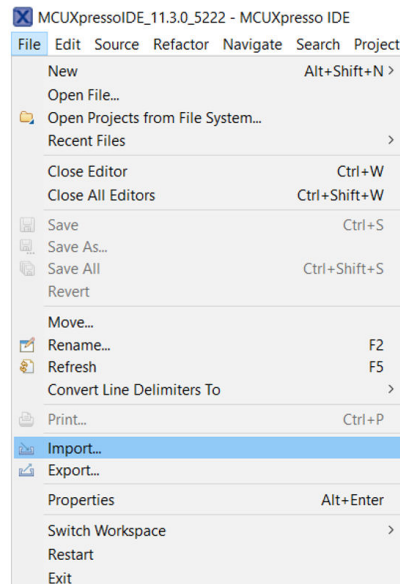


Figure 5. Selecting File->Import in MCUXpresso

3. In the resulting dialog box, select **General-> Existing Projects into Workspace** and click **Next**.

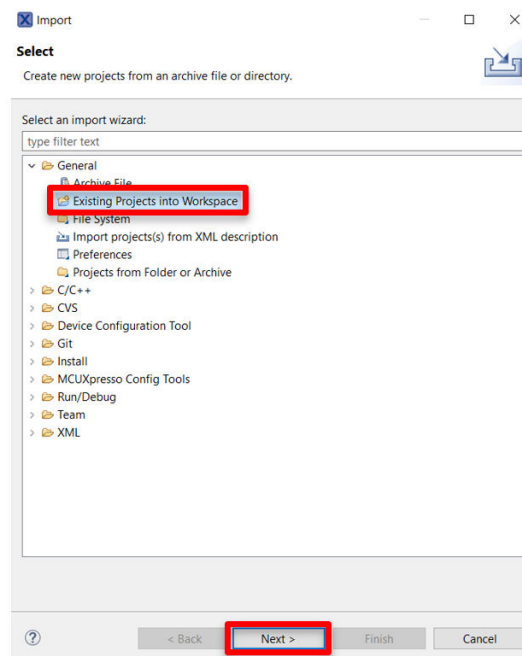


Figure 6. Import dialog box

4. Then point MCUXpresso IDE to the location where the Smart Kitchen Demo software was unzipped. Click **Finish** to import the project.

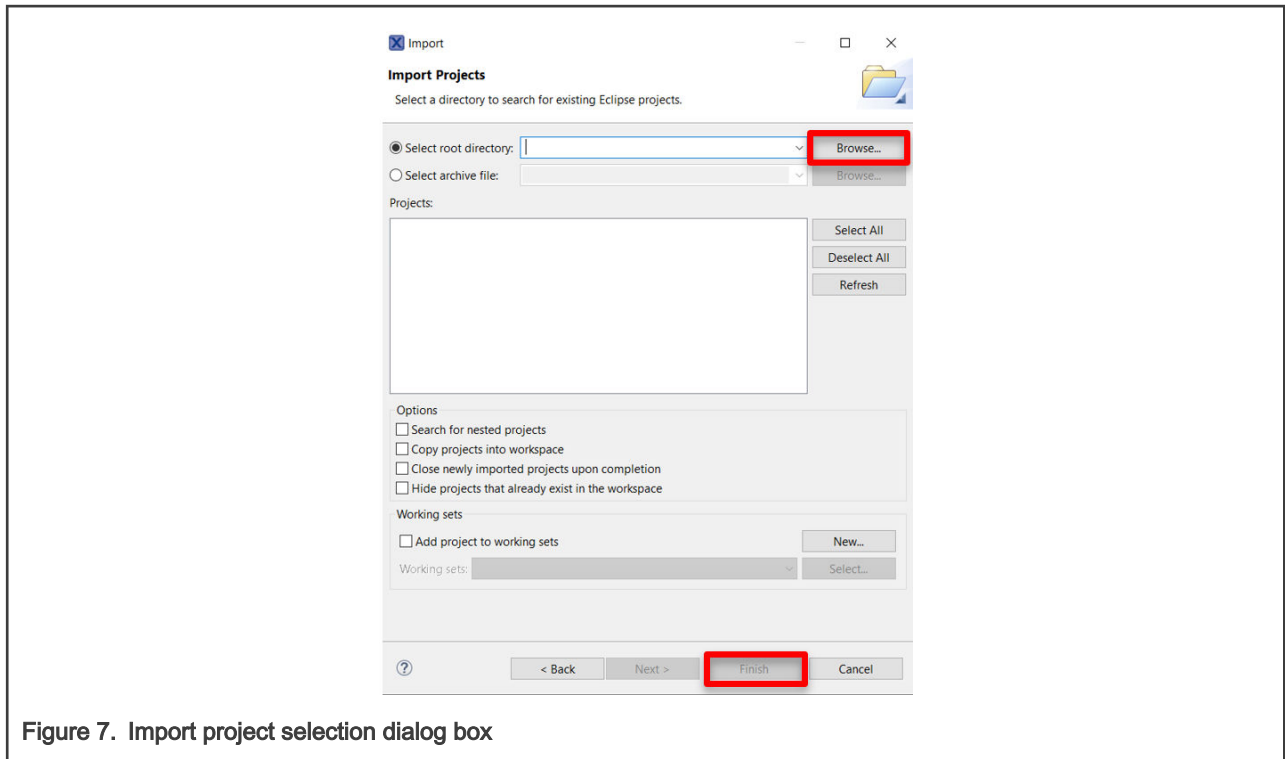


Figure 7. Import project selection dialog box

2.1.2 Load the project to the board

To load the project to the board, click the **Build** icon to build the project.



Figure 8. Build icon

With the MIMXRT1050-EVKB evaluation kit connected to the PC, click the **Debug** icon to load the project.



Figure 9. Debug icon

2.1.3 Loading the GUI Guider project

1. Open the GUI Guider application.
2. Upon opening GUI Guider, the Create New Project screen is displayed.

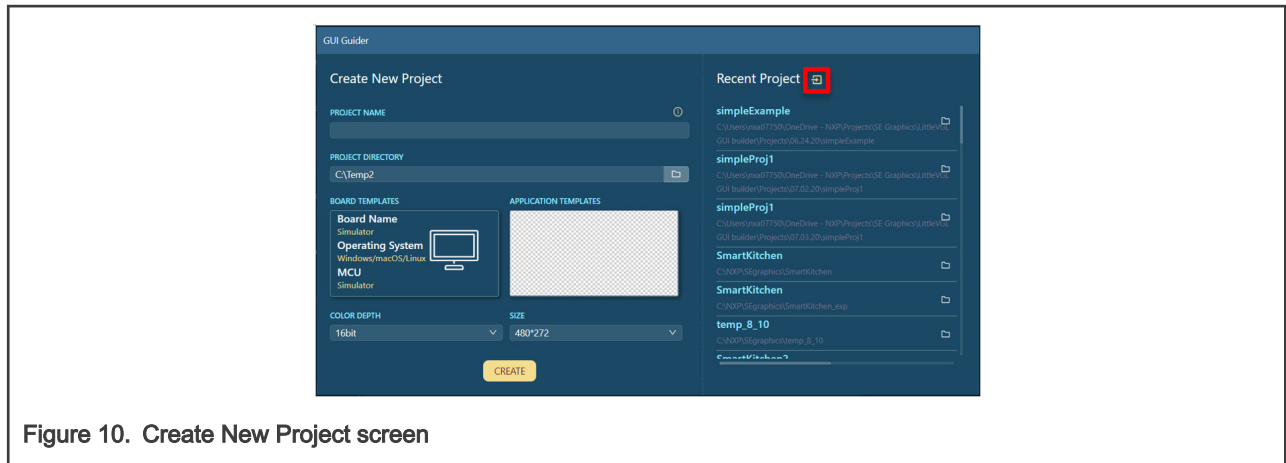


Figure 10. Create New Project screen

- From the Create New Project screen, select the **Import project** button to point GUI Guider to the location of the project:
`<Smart Kitchen Demo Software root>/ smart_kitchen_demo_gg.`

2.2 Smart Kitchen Demo operation

When the demo begins, the default screen is displayed allowing you to select one of three demos.

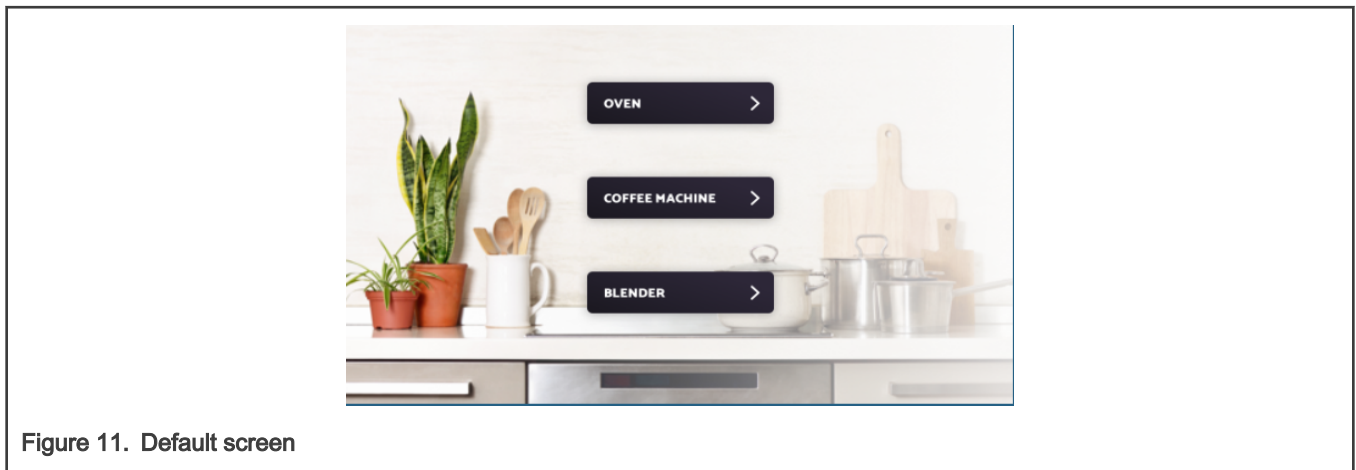


Figure 11. Default screen

To load one of the demos, touch the button of the desired demo on the default screen.

2.2.1 Oven demo

The flow of the Oven demo portion is as follows:

Date/time screen -> Oven settings -> Oven Cook

When the Oven demo is first loaded, you see the Oven home screen (shown below).

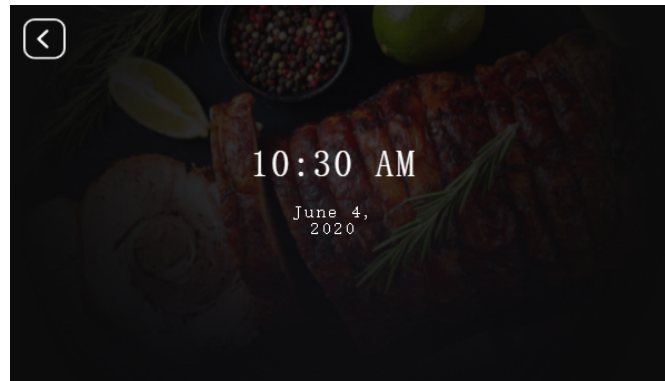


Figure 12. Oven home screen

The Oven home screen displays the time and date and includes a **Back** button to navigate back to the default screen. To move from the Oven home screen to the settings screen, touch the screen anywhere other than the **Back** button.

2.2.1.1 Oven settings screen

After you touch the home screen, the Oven settings screen is loaded. This screen is shown in the following figure.

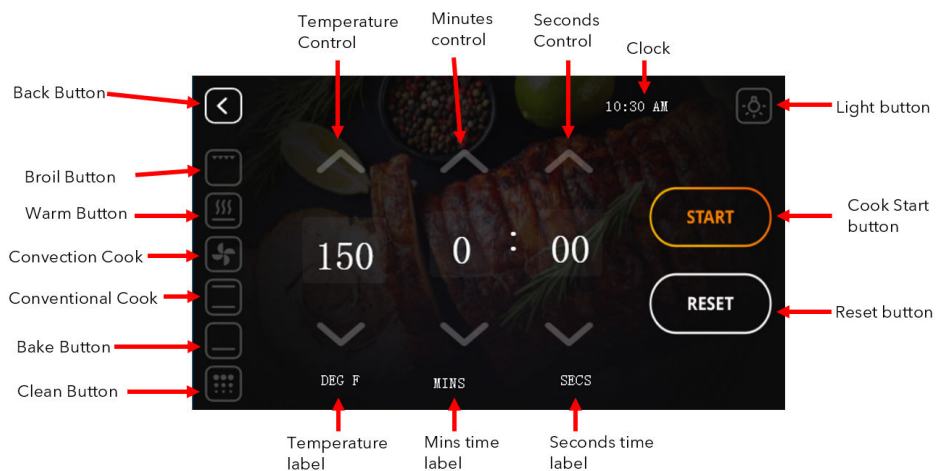


Figure 13. Oven settings screen

On this screen, there are the following functions:

- Select the oven mode
- Set the oven temperature
- Configure the cooking time
- Reset the oven settings (cooking time, oven mode, and oven temperature)
- Turn on/off the oven light
- Navigate back to the start screen (default screen)
- Start cooking
- Clock

2.2.1.1.1 Selecting the Oven mode

To select the Oven mode, press one of the Oven mode buttons from the left-hand side of the settings screen.

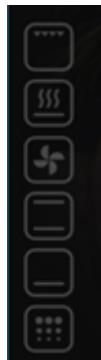


Figure 14. Oven mode buttons

These buttons are toggle buttons. Once pressed, they retain their state (the checked state) until they are pressed again to release the state (unchecked state). Only one of these buttons may be in the checked state at a time. If you press an unchecked button while there is another button already in the checked state, the selected button automatically reverts to the unchecked state.

Before cooking is allowed to start, put one of the mode buttons into the checked state.

2.2.1.1.2 Setting the Oven temperature

You can set the oven temperature using the Oven temperature control.

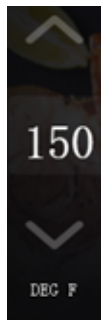


Figure 15. Oven temperature control

The Oven temperature control consists of a roller, an up arrow, and a down arrow. You can change the cooking temperature of the oven by swiping up or down on the number, or by using the up/down arrow controls.

Possible temperatures range from 150°F to 400°F in increments of 5°F.

2.2.1.1.3 Configuring the cooking time

You can configure the cooking time of the oven using the cooking time controls shown below.

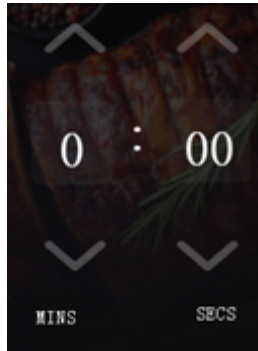


Figure 16. Cooking time control

The cooking time control consists of two rollers, two up arrows, and two down arrows. Set the minutes to cooking by the roller and up/down arrow on the left and the seconds to cook by the roller and up/down arrow on the right. Set both the minutes and seconds of the cooking time in a similar manner to the oven temperature. You can either swipe up/down on the numbers or use the up/down arrow to change the time one unit at a time. You can configure a cooking time of up to 8:59.

Before cooking, select a valid time (more than 0:00).

2.2.1.1.4 Resetting the Oven Settings

You can reset the Oven settings by pressing the reset button, shown below.



Figure 17. Reset button

This button clears the oven mode (all oven mode buttons are in the unchecked state), sets the temperature to 150°F, and sets the cooking time to 0:00.

2.2.1.1.5 Oven light

The **Oven light** is a toggle button on the upper right-hand side of the screen. This button is shown below.



Figure 18. Oven light button

When this button is inactive (as shown), the indicator is off. When selected, the button turns orange and the indicator lights up.

2.2.1.1.6 Back button

To get back to the default (start) screen, press the **Back** button (shown below).



Figure 19. Back button

2.2.1.1.7 Start cooking

To start cooking, press the **Start** button (shown below) when all inputs are valid.



Figure 20. Start button

The **Start** button checks that the inputs are valid and then loads the Oven cook screen. Pressing the Start button with an invalid input results in the Start button press being ignored. The valid inputs are as follows:

- A cooking mode must be in the checked state
- A valid time (more than 0:00) must be configured

2.2.1.2 Oven cook screen

The oven cook screen shows the operation of the oven when cooking. It is composed of a temperature indicator, a timer, a **Cancel** button, a **Light** button, a clock, and a **Back** button.

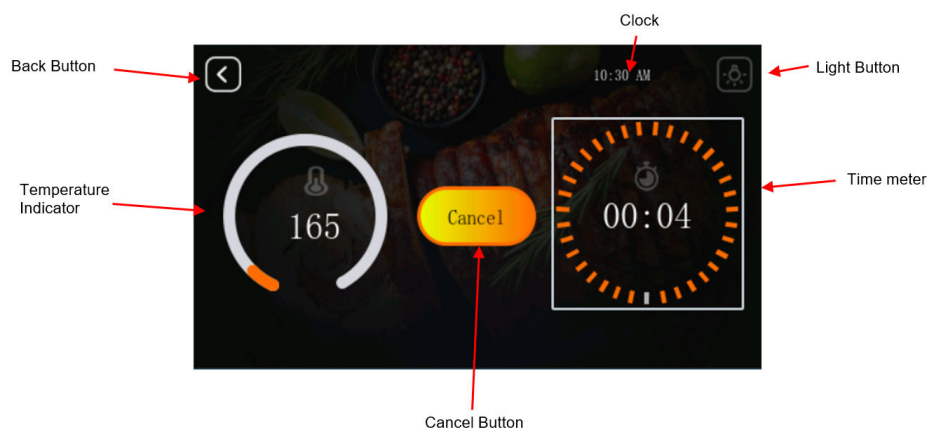


Figure 21. Oven cook screen

2.2.1.2.1 Temperature indicator

The Temperature indicator is composed of an Arc widget, a label widget, and an image widget.

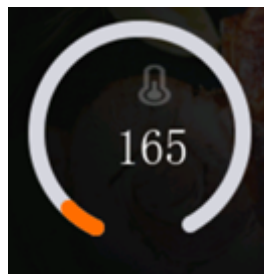


Figure 22. Temperature indicator

The text for the label as well as the value of the Arc widget, are obtained from the temperature roller in the Oven settings screen. This widget requires no user interaction.

2.2.1.2.2 Time meter

The time meter is composed of a line meter widget, a label widget, and an image widget.

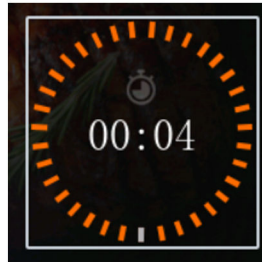


Figure 23. Time meter

The label decrements and the time meter value decrements as well. This widget also requires no user interaction.

2.2.1.2.3 Oven light

The **Oven light** is a toggle button on the upper right-hand side of the screen. This button is shown below.



Figure 24. Oven light button

When this button is inactive (as shown), the indicator is off. When selected, the button turns orange and the indicator lights up.

2.2.1.2.4 Back button

To get back to the Oven settings screen, press the **Back** button (shown below).



Figure 25. Back button

2.2.1.2.5 Cancel button

You can reset the cooking time by pressing the **Cancel** button, shown below.

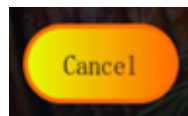


Figure 26. Cancel button

This button stops the timer and resets the cooking time to 0:00. To get back to the settings screen to restart the cooking timer, press the **Back** button.

2.2.2 Coffee machine demo

The flow of the coffee machine portion is as follows:

Café Home screen → Café settings → Café Brew Screen → Café Settings

2.2.2.1 Café home screen

The Café home screen is shown below.

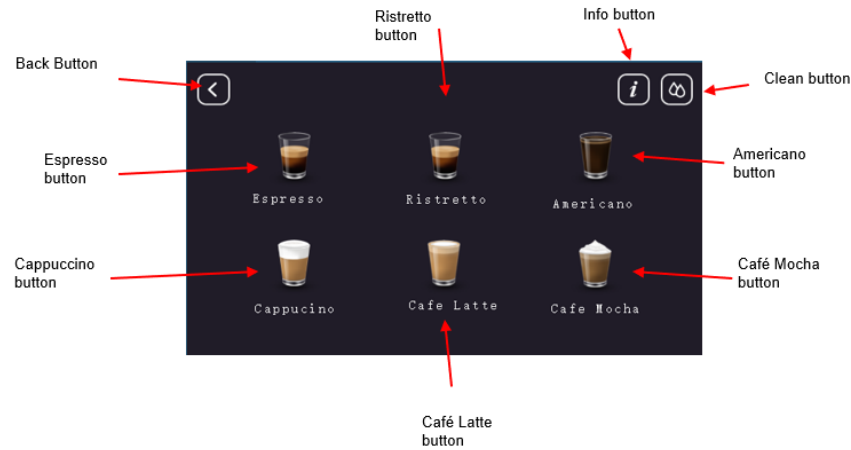


Figure 27. Café home screen

The home screen allows you to select between six different flavors of coffee. It is composed of six image buttons and six labels that make up the coffee selections. It also has the **Back** button, the **Information** button, and the **Cleaning mode** button.

2.2.2.1.1 Back button

Pressing the **Back** button from this screen loads the Default (start) screen.



Figure 28. Back button

2.2.2.1.2 Coffee selection

Pressing any one of the six coffee cup image buttons loads the Café settings screen that allows more settings to be selected.

2.2.2.2 Café settings screen

The Café settings screen is shown below.



Figure 29. Café settings screen

Once the Café Brew settings screen is loaded, the selected flavor is displayed in the middle of the screen. From this screen, select the brew strength, the cup size, and start the brewing/filling process.

2.2.2.2.1 Selecting the cup size

Select the cup size using the cup size selections on the left-hand side of the screen (shown below).

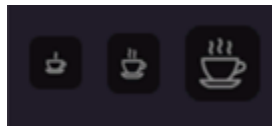


Figure 30. Cup size buttons

Select only one cup size at a time. If you select another cup size when one is already selected, the previous cup size is de-selected and the new cup size is selected. Selected cup sizes are highlighted in white, while de-selected cup sizes are in gray.

2.2.2.2.2 Setting the strength

The strength of the drink is set using the slider on the right-hand side of the screen. This slider is shown below.

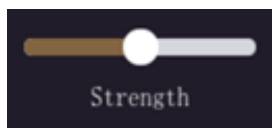


Figure 31. Strength slider

To increase the drink strength, slide the slider knob to the right. To decrease the drink strength, slide the slider knob to the left.

2.2.2.2.3 Returning to the previous screen

You can return to the drink selection screen using the **Back** button (shown below) in the upper left-hand corner of the screen.



Figure 32. Back button

2.2.2.2.4 Beginning the brew/fill

You can start the brew/fill process by pressing the **Start** button (shown below) in the center of the screen.

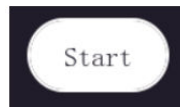


Figure 33. Start button

You cannot start the brew/fill process without selecting a cup size.

2.2.2.3 Café brew screen

Once the strength and cup size are set on the setting screen, press the **Start** button and the Café Brew screen is loaded.

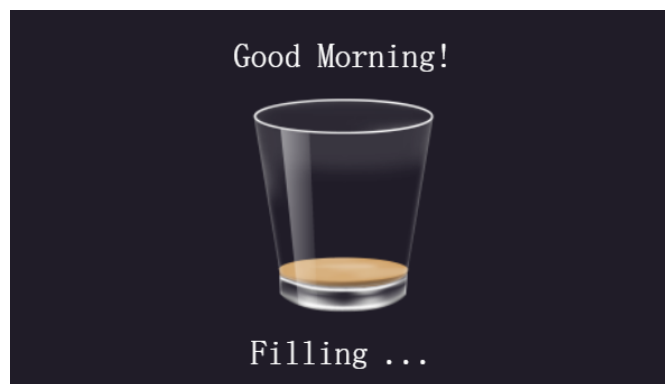


Figure 34. Café Brew screen

This screen requires no user interaction. It displays a "Good Morning" message and status message below the cup image. The cup image animates displaying the status of the fill process. Once finished, the "Good Morning!" text changes to "Done!" and the Cafe Home screen is automatically displayed allowing for the selection of a new cup of coffee.

2.2.3 Blender demo

The Blender demo has only one screen associated with it. The blender screen is shown below.

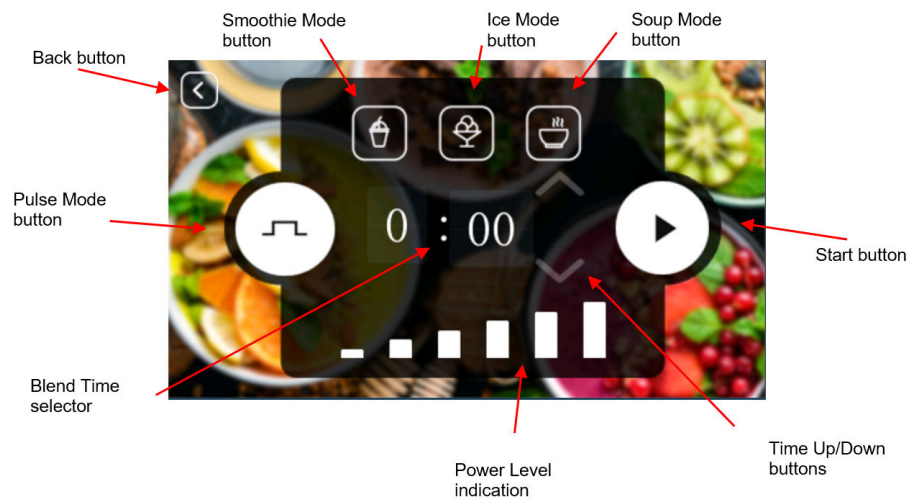


Figure 35. Blender screen

From the blender screen, select one of the four modes: pulse, smoothie, ice, soup. Selection of one of these modes automatically sets a default power level and blend time. You can change the blend time and power level after it is set.

2.2.3.1 Selecting the blender mode

Select the blender mode by pressing any one of the four mode buttons.

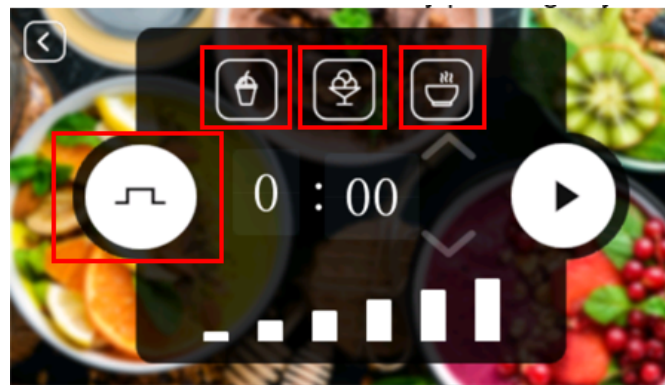


Figure 36. Blender mode selections

Select only one mode at a time. If you select a mode while a different mode is already selected, the currently selected mode is deselected and the new mode is selected. Unselected modes appear in gray, while selected modes are colored in pink (except for the pulse mode button, which is gray when selected instead of its normal white).

When you select a mode, the power level and blend time are updated.

2.2.3.2 Changing the blend time

Adjust the blend time using the time adjustment controls in the center of the screen.



Figure 37. Time adjustment controls

Change the time by swiping up or down on either the minute place or the second place. You can also press the up/down arrows to the right to increment/decrement the current time by one second. A maximum of 9:59 is possible.

2.2.3.3 Adjusting the power level

The power level is indicated by the power level graph at the bottom of the screen.



Figure 38. Power level graph

The power level is indicated by pink graph bars. The more bars are pink, the more powerful the setting is. Cycle through different power levels by pressing the graph.

2.2.3.4 Starting the blender

Start the blender by pressing the **Start** button (shown below).



Figure 39. Start button

To start the blending process, configure a valid power level (at least one pink bar), time (at least 0:01 or more), and blender mode.

2.2.3.5 Returning to the default screen

Return to the default (start) screen by pressing the **Back** button (shown below).



Figure 40. Back button

3 Smart Kitchen demo source code

This section provides a general overview of the file structure and ways of modifying the Smart Kitchen demo.

3.1 File structure

The Smart Kitchen Demo is an example for the i.MXRT1050 EVKB and is based on the `littlevgl_guider` demo in the i.MXRT1050 SDK v2.9.0. All necessary code and project files are in the main folder. The main source code folder is shown below.

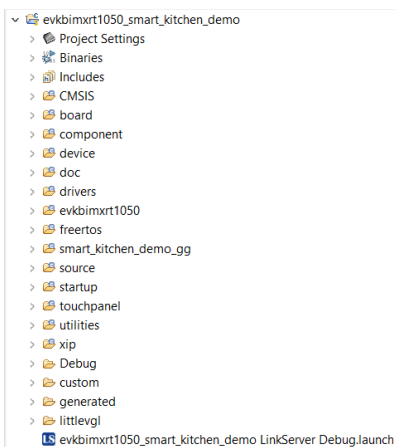


Figure 41. Smart Kitchen Demo

The folder structure follows the standard NXP example project folder structure and the GUI Guider project is in the `smart_kitchen_demo_gg` folder. It allows changes to the GUI Guider project without any manual copy and paste of files or folders. It also allows the GUI Guider project to be transported easily when sharing the project among colleagues or customers.

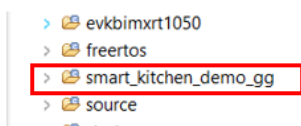


Figure 42. GUI Guider folder for Smart Kitchen Demo

Not all files under this folder are used by the project. Only the necessary files are included in the build. Including other files can cause build issues.

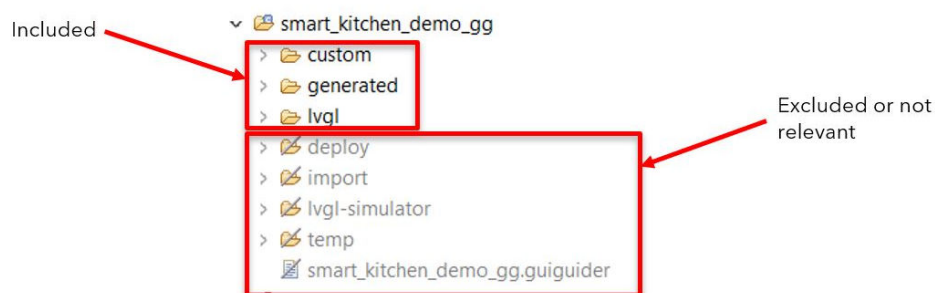


Figure 43. Included / Excluded GUI Guider files for Smart Kitchen Demo

Important folders are the `custom` folder, `generated` folder, and `LVGL` folder. These folders are discussed below.

The `custom` folder contains `custom.c` and `custom.h`. These files hold custom code that the GUI Guider project files may need to run. They are not modified when rebuilding the GUI Guider project. These files are the link between the GUI Guider project and the rest of the application.

The `generated` folder contains all files generated by the GUI Guider. These files are modified by the GUI Guider when rebuilding the GUI Guider project. They must not be modified manually.

The `LVGL` folder contains the LVGL library source files used by the GUI Guider source files.

3.2 Modifying the Smart Kitchen Demo

There are two ways to modify the Smart Kitchen Demo:

1. Via the GUI Guider
2. Via the source code directly

The first way (via the GUI Guider) is the preferred method of modifying the Smart Kitchen Demo as it is the cleanest and most seamless. Some situations, however, require the source code to be modified. Typically it is required when a function that GUI Guider cannot provide is needed (such as a GUI command initiated from an MCU peripheral). Both of these methods are discussed here.

3.2.1 Modifying the demo via GUI Guider

The full GUI Guider project allows the demo to be modified. The project is located in `<root>/smart_kitchen_demo_gg/smart_kitchen_demo_gg.guiguider`. To open the project, open the GUI Guider and click the **Import Project** button.

If the GUI Guider is already open, click **File->Open** and point the dialog box to the project location.

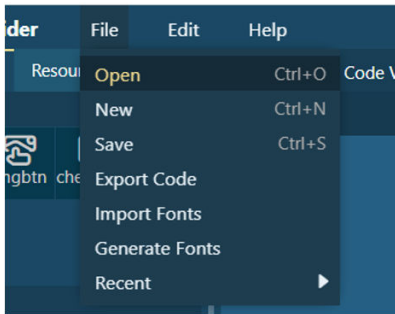


Figure 44. Opening a GUI Guider project from GUI Guider with File->Open

Once the project is open, you can add, delete, or modify widgets as per normal operation. The changes are seamless and it is not required to export the code to a specific location to incorporate the changes. Generate the code and rebuild and download the application to the target.

3.2.2 Modifying the demo via the source code

When modifying the demo via direct source code modifications, be aware of certain files that should not be modified because the GUI Guider modifies them. Files in the `smart_kitchen_demo_gg` folder, except for the custom folder, must not be modified directly.

You can modify files in the `smart_kitchen_demo_gg/custom` folder directly. Inside this folder, there are three files: `custom.c`, `custom.h`, and `custom.mk`. Modifications must be made to `custom.c` and `custom.h`.

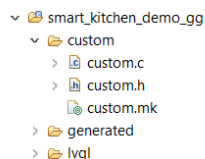


Figure 45. `Smart_kitchen_demo_gg/custom` folder

When the simulator runs, functions from `custom.c` and `custom.h` are compiled and included in the simulation. So any code that you want to be simulated in your modifications must be included in `custom.c` and `custom.h`. Do not include the code that is MCU-specific as the simulator may not be able to compile or run this code.

4 Tips and tricks

This section shares tips and tricks used in this application.

NOTE

This application is based on the v1.0 release of the GUI Guider.

4.1 Setting default values before a screen is created/loaded

When you launch the GUI Guider application, only the default screen is loaded. It means that no other screens and their widgets are created for applications with multiple screens. If your application must modify widgets on these screens before loading them, a runtime error occurs if your application cannot solve this problem. As an example, the Café Settings screen is examined.

The Café Settings screen is loaded from the Café Home screen. On the Café Home screen, you can select desired coffee flavor.

This flavor is displayed on the Café Settings screen and the text above the coffee cup is also updated accordingly.

Changing the coffee cup image and associated text is a runtime action that takes place before the screen loads. Not updating these widgets before the screen loads results in undesirable blips/flickers that degrade the user experience. As an example, consider the Espresso button event handler shown below.

```

153 static void CafeHm_espressoBtnevent_handler(lv_obj_t * obj, lv_event_t event)
154 {
155     switch (event)
156     {
157     case LV_EVENT_RELEASED:
158     {
159         lv_obj_clean(lv_scr_act());
160         setup_scr_CafeSettings(&guider_ui);
161         lv_scr_load(guider_ui.CafeSettings);
162         cafeBrewBtn_cb(obj, event);
163     }
164     break;
165     default:
166     break;
167     }
168 }

```

Figure 46. Espresso button event handler

When you release the **Espresso** button, the `LV_EVENT_RELEASED` case is executed in the `CafeHm_espressoBtnevent_handler` function. This function clears the current screen, then calls the `setup_scr_CafeSettings` function that creates and configures the widgets on the Café Settings screen. Next, the `lv_scr_load` function is called. This function loads the screen into the LVGL kernel. After that, the `lv_task_handler` is called. Since it happens in the event handler, it is called from an interrupt. Therefore, it should not take effect immediately. The changes occur after this function exits and `lv_task_handler` is called again. It enables the final function `cafeBrewBtn_cb` that configures the coffee cup image and associated text.

5 Conclusion

In this application note, it has been shown how to operate the Smart Kitchen Demo developed for i.MXRT1050 using GUI Guider and LVGL. Additionally, the source code has been examined and a description of how to modify the source has been outlined. It provides a helpful jumping-off point for the application development.

For additional information regarding the GUI Guider, see www.nxp.com/gui-guider. If you have questions regarding this demo or need support, submit a question to the NXP community at community.nxp.com.

6 Revision history

Table 1. Revision history

Revision number	Date	Substantive changes
0	18 October 2021	Initial release

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